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Question Answering over Geospatial Knowledge Graphs

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Outline

- Fundamentals
- Knowledge Graph Question Answering
- Geospatial Question Answering
- Geospatial Question Answering Datasets
- Geospatial Question Answering Systems
- Evaluation
- Geospatial Agentic Systems

The slide features a decorative header at the top with a blue background and an orange wavy line. On the left and right sides, there are vertical lines in blue and orange, some with small dots, resembling circuit traces.

Basic Concepts

Basic Concepts

Question Answering

- **Question Answering** is the Computer Science discipline of understanding and responding to questions made in natural language.
- Core use of chatbots like **ChatGPT** and **Gemini**.
- Knowledge is “stored” in model weights.

When was George Washington born?

George Washington was born on **February 22, 1732** (according to the modern Gregorian calendar).

At the time of his birth, the British colonies still used the **Julian calendar**, so his birthday was recorded then as **February 11, 1731** — but after the calendar change in 1752, the date shifted to February 22, 1732.



and where did he grow up?

George Washington grew up in the **British colony of Virginia**, primarily along the **Potomac River**.

He spent much of his childhood at several family properties, including:

- **Pope's Creek (Wakefield)** in **Westmoreland County, Virginia** – his birthplace and early childhood home.
- **Ferry Farm**, near **Fredericksburg, Virginia** – where he lived from about age 6 to his late teens.
- Later, he inherited and expanded **Mount Vernon**, located on the Potomac River, which became his lifelong residence and plantation.

So, in short — Washington grew up in **rural colonial Virginia**, surrounded by farmland, rivers, and the plantation life typical of the Virginia gentry.

Basic Concepts

LLM Hallucinations

- Knowledge is stored in model weights.
 - not really knowledge!
 - requires retraining to update!
- LLMs are prone to hallucinations.
- **Hallucination** is a response generated by AI that contains false or misleading information presented as fact.

How many people live in Pikermi?

Assistant B

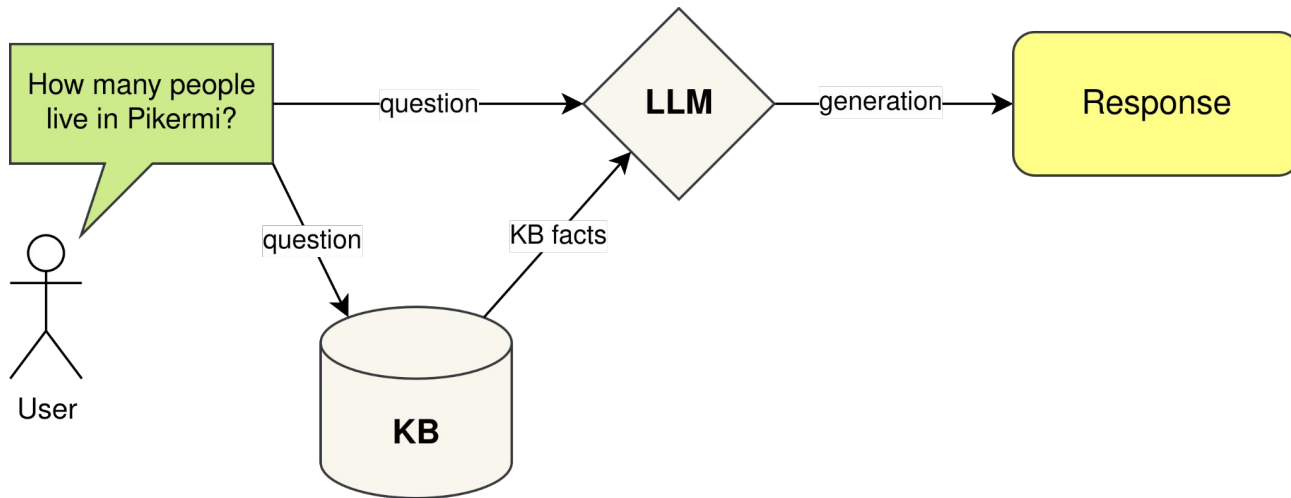
As of the most recent data, Pikermi, a community in the northern part of Cyprus, has a population of approximately 2,000 to 2,500 people. However, for the most accurate and up-to-date statistics, it's best to consult the latest census or local government sources.

Country	Greece
Administrative region	Attica
Regional unit	East Attica
Municipality	Rafina-Pikermi
Area	
• Municipal unit	21.522 km ² (8.310 sq mi)
Elevation	140 m (460 ft)
Population (2021)^[1]	
• Municipal unit	7,707
• Municipal unit density	358.1/km ² (927.5/ sq mi)

Basic Concepts

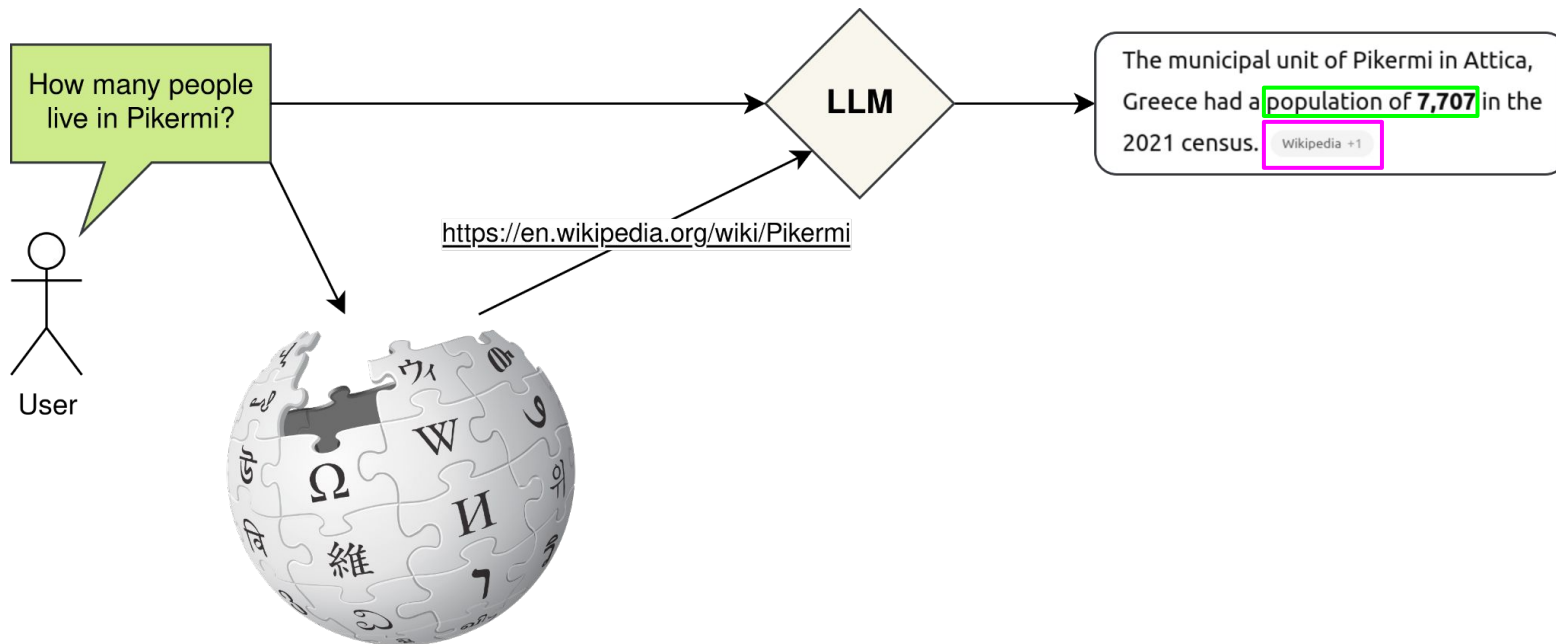
Reducing Hallucinations - Retrieval Augmented Generation

- Turns out that LLMs have clear weaknesses when generating answers from scratch.
- Much better at **information extraction** and **synthesis**.
- **Retrieval Augmented Generation** is a technique where relevant facts from an external knowledge base are given to the LLM as context when deemed necessary.



Basic Concepts

Reducing Hallucinations - Retrieval Augmented Generation



Basic Concepts

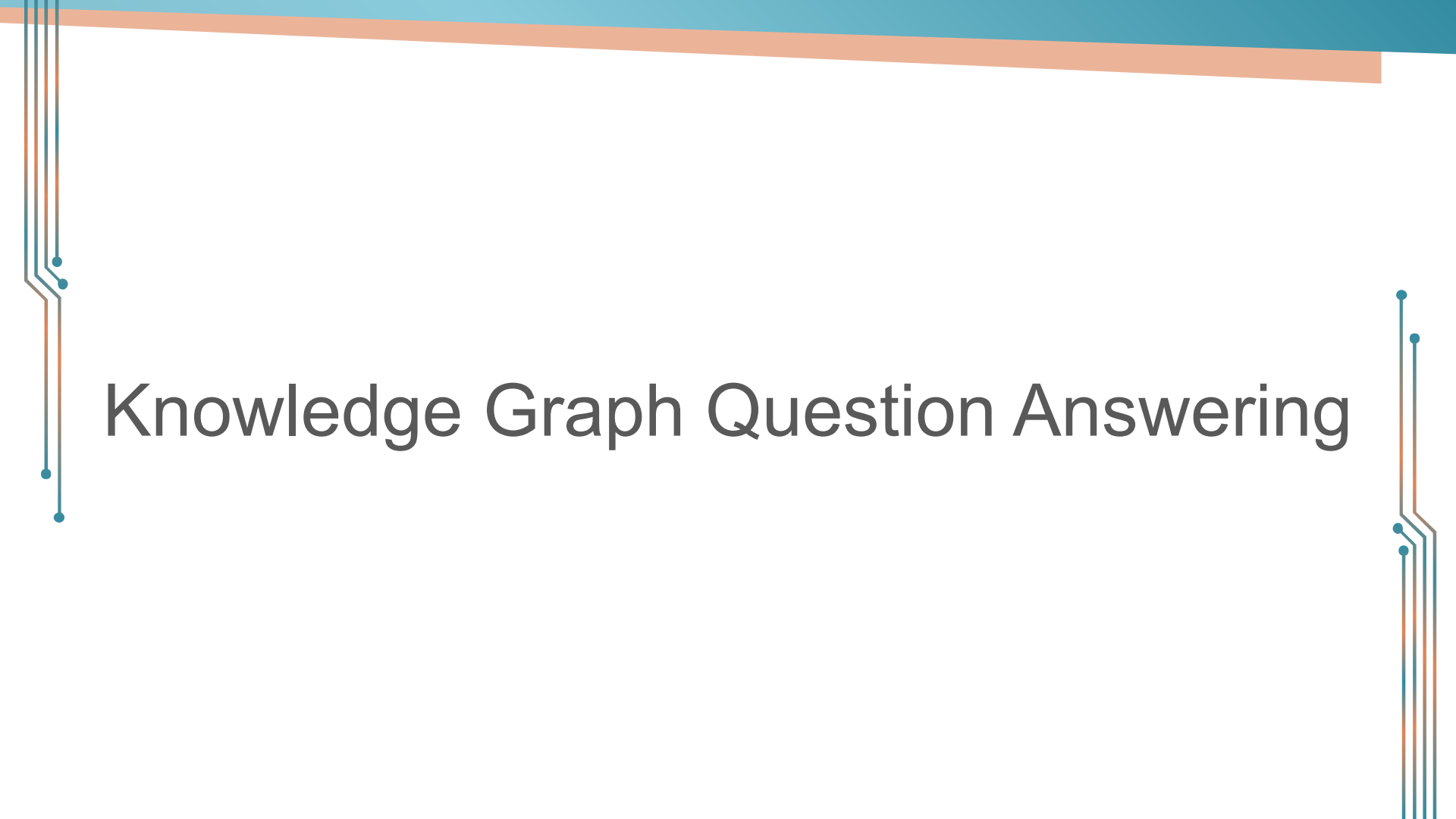
Retrieval Augmented Generation - Information Sources

- Documents, Web pages
 - Simplest and most often used form of RAG. Facts can be coarse or fine grained.
 - Remember: LLMs are very good at **information extraction** and **synthesis**.
- Tabular Data
- Relational Databases
- Knowledge Graphs (GraphRAG)
 - **Relationships**
- Multimodal (images and audio with captions)

Basic Concepts

GraphRAG & Knowledge Graph Question Answering

- **GraphRAG** is **RAG** that utilizes a **Knowledge Graph** as its information source.
 - KG is used to retrieve context to enhance LLM reasoning + provide facts
 - KG might not be pre-existing, but constructed by the system to keep track of relations
 - RAG is applied if deemed necessary
 - Used when relations between entities are important to the task at hand (multi-hop reasoning)
- **Knowledge Graph Question Answering** is the **Question Answering** task that answers questions over a **Knowledge Graph**.
 - Predates RAG and LLMs
 - Natural language interface to a KG, answers are always extracted from the KG, usually through formal queries (SPARQL)
 - Useful to both humans...
 - novice SPARQL users
 - expert SPARQL users who are interacting with an unknown ontology
 - .. and LLMs
 - is used as a retriever component for GraphRAG!



Knowledge Graph Question Answering

Knowledge Graph Question Answering

Process Overview

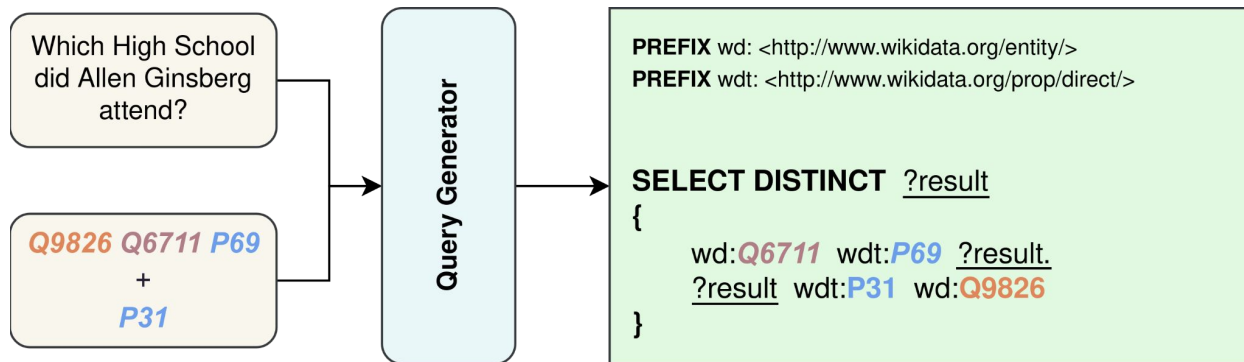
Knowledge Graph Question Answering (KGQA) systems generally include two steps:

- **Entity linking** is the process of grounding the input question to the target KG.
 - **Named Entity Recognition and Disambiguation**
 - **Class Linking**
 - **Property Linking**

Which **high school** did **Allen Ginsberg** **attend** ?

Q9826 Q6711 P69

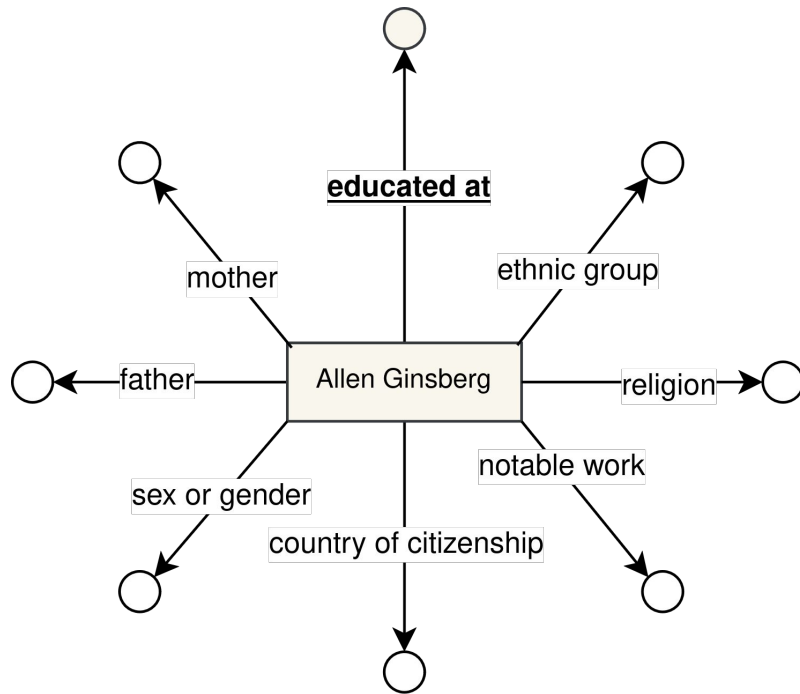
- **Answer Extraction** is the process of extracting the requested information from the KG (traditionally via a formal query).

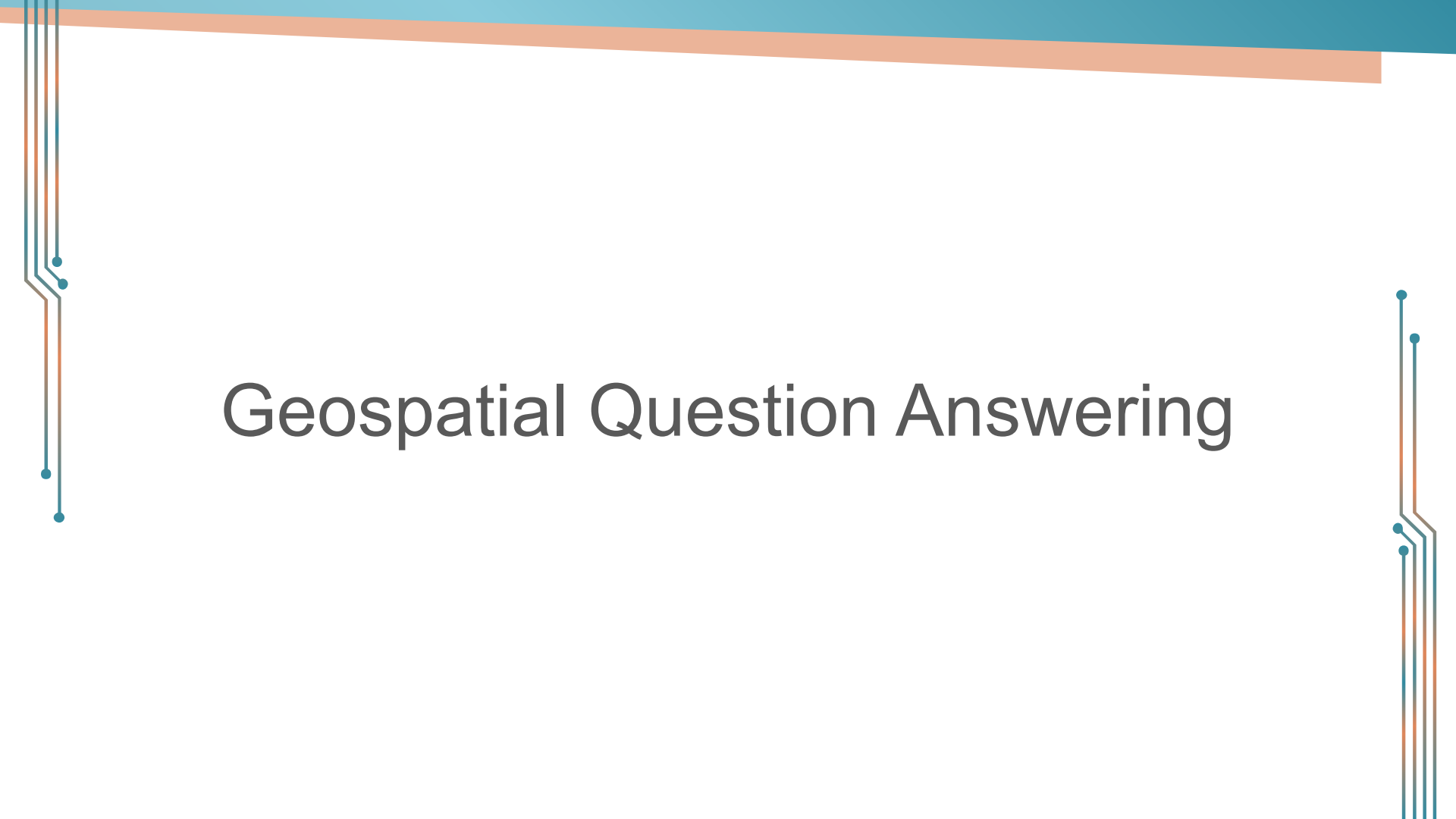


Knowledge Graph Question Answering

Answer Extraction

- **Semantic Parsing**, natural language questions are translated to executable queries (**Text-to-SPARQL**)
 - Most widely used technique. Most relevant to the geospatial domain (we will see why).
 - Categories of systems:
 - Template based query generation
 - seq2seq models
 - Large Language Models
 - The query generation / template selection step usually happens after entity linking, but can also proceed it.
- **Information Retrieval & Graph Reasoning**, models extract information from the KG either through embeddings or through stepwise exploration
 - Graph reasoning techniques have seen a lot of growth in the last few years.
 - Particularly iterative exploration of the KG, starting from linked entities.





Geospatial Question Answering

Geospatial Question Answering

Definition

Geospatial Questions require qualitative or quantitative geospatial knowledge to be answered.

- **qualitative**, e.g., “Which states does the Murray River cross?”
- **quantitative**, e.g., “What is the distance between Brisbane and Sydney?”

Depending on the complexity these can be **answered directly** by an LLM ...

Which states does the Murray River cross?

The **Murray River**, Australia’s longest river, crosses and forms borders between several states:

- **New South Wales (NSW)** – The Murray flows along much of the southern border of NSW.
- **Victoria** – It forms the **northern border** of Victoria, separating it from NSW.
- **South Australia (SA)** – The river then flows **into South Australia**, continuing westward before emptying into the **Southern Ocean** near Lake Alexandrina.

So, in summary, the Murray River crosses or borders **New South Wales, Victoria, and South Australia**.

Geospatial Question Answering

RAG

Geospatial Questions require qualitative or quantitative geospatial knowledge to be answered.

- **qualitative**, e.g., “Which states does the Murray River cross?”
- **quantitative**, e.g., “What is the distance between Brisbane and Sydney?”

Depending on the complexity these can **require RAG on documents** when available ...

What is the length of the largest motorway in Germany?

The longest motorway in Germany is Bundesautobahn 7 (A 7), with a length of approximately **963 km**.

en.wikipedia.org +3

Geospatial Question Answering

Problem

Geospatial Questions require qualitative or quantitative geospatial knowledge to be answered.

- **qualitative**, e.g., “Which states does the Murray River cross?”
- **quantitative**, e.g., “What is the distance between Brisbane and Sydney?”

Depending on the complexity these can **require spatial operations** via the use of **external spatial knowledge bases** and **spatial tools**.

How many villages in Japan are more than 20kms away from the nearest hospital and have more than 1000 inhabitants?

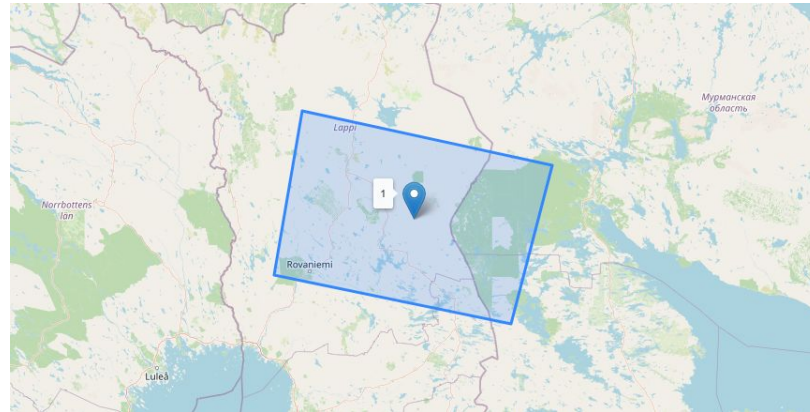


Determining the exact number of villages in Japan that are more than 20 kilometers away from the nearest hospital and have populations exceeding 1,000 is challenging due to the lack of comprehensive, publicly available data that combines both geographic distance and population size criteria.

Geospatial Question Answering

Geospatial Knowledge

- **Geospatial Knowledge** is represented as:
 - coordinate pairs
 - detailed geometries (Points, Lines, Polygons)
- Geospatial knowledge is often organized in **GeoKGs** → external spatial knowledge bases
- These are queried via an extension of SPARQL known as **GeoSPARQL** → spatial tools



A point and a geometry on a map.

Geospatial Question Answering

Geospatial Operations - GeoSPARQL

GeoSPARQL extends SPARQL to support geospatial queries through a number of spatial functions.

Question: Which **nature reserves** in **Ireland** have **lakes** within their boundaries?

Equivalent GeoSPARQL over YAGO2geo:

SELECT DISTINCT ?p1 WHERE

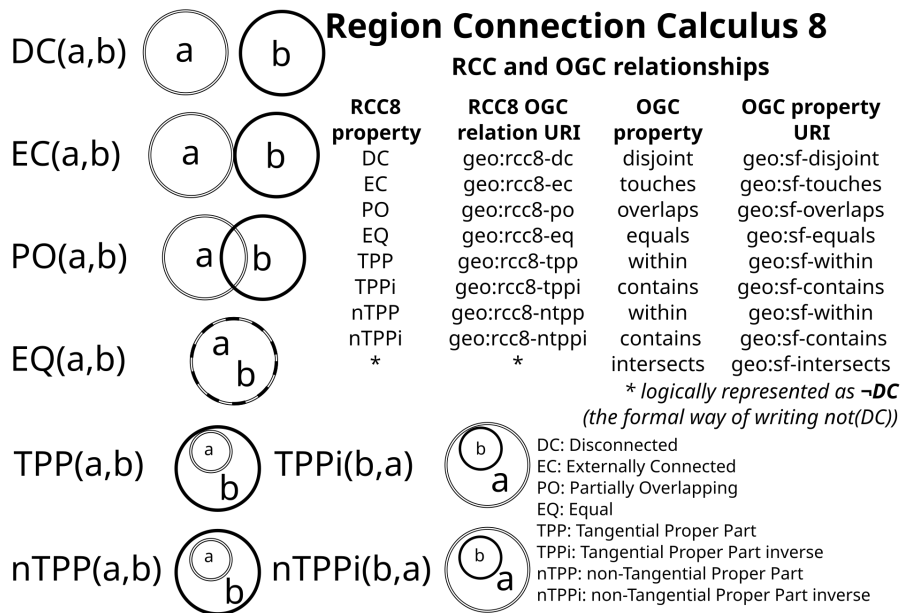
```
{
  ?p1 rdf:type y2geo:OSM_nature_reserve ;
      geo:hasGeometry ?p1geo .
  ?p1geo geo:asWKT ?p1WKT .
  ?p2 rdf:type y2geo:OSM_lake ;
      geo:hasGeometry ?p2geo .
  ?p2geo geo:asWKT ?p2WKT .

  yago:Republic_of_Ireland geo:hasGeometry ?o1 .
  ?o1 geo:asWKT ?iWKT .
```

FILTER(geof:sfWithin(?p1WKT, ?iWKT))

FILTER(geof:sfContains(?p1WKT, ?p2WKT))

}



Geospatial Question Answering

Geospatial Operations - SQL

Similar standards exist for relational databases, *OGC Simple Features*.

Question: Which nature reserves in Ireland have lakes within their boundaries?

Equivalent GeoSPARQL over YAGO2geo:

SELECT DISTINCT ?p1 **WHERE**

```
{
  ?p1 rdf:type y2geo:OSM_nature_reserve ;
      geo:hasGeometry ?p1geo .
  ?p1geo geo:asWKT ?p1WKT .
  ?p2 rdf:type y2geo:OSM_lake ;
      geo:hasGeometry ?p2geo .
  ?p2geo geo:asWKT ?p2WKT .

  yago:Republic_of_Ireland geo:hasGeometry ?o1 .
  ?o1 geo:asWKT ?iWKT .

  FILTER(geof:sfWithin(?p1WKT, ?iWKT))
  FILTER(geof:sfContains(?p1WKT, ?p2WKT))
}
```

Equivalent SQL:

```
SELECT DISTINCT nr.id AS p1
FROM nature_reserve nr, lake l, country irl
WHERE irl.name = 'Republic_of_Ireland'
      AND ST_Contains(nr.geom, l.geom)
      AND ST_Within(nr.geom, irl.geom);
```

Geospatial Question Answering

Why Semantic Parsing for GeoQA?

Semantic Parsing, natural language questions are translated to executable queries (**Text-to-SPARQL**)

- Advantages:
 - Seamlessly combines question-answering with spatial operations.
 - Takes advantage of decade-long optimizations in triplestores (and RDBMS).
- Enables us to efficiently execute GeoQA queries over very large databases.
 - With some additional optimizations that we will also present in this tutorial!

Information Retrieval & Graph Reasoning systems on the other hand need to be expanded with external tools to conduct spatial operations.

- Disadvantages:
 - These tools are often geospatial triplestores and/or RDBMS (which means that we return to semantic parsing).
 - If spatial operations are implemented through simpler spatial functions, efficiency can start to be a problem.

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GeoQA Datasets

GeoQA Datasets

Geospatial Datasets - GeoQuestions201

GeoQuestions201 [1] contains **201** question-query-answer triples. Queries are written in **GeoSPARQL**.

- Questions crowdsourced from students
- Target knowledge graph created by interlinking:
 - **DBpedia** [2]
 - **OpenStreetMap** [3] (UK and Ireland)
 - **Global Administrative Areas** [4] (UK and Ireland)
- Wide variety of questions organized in **9 categories**
- Has been used as a gold standard to test the effectiveness of GeoQA [1] and other geospatial QA engines

[1] Punjani, D., Singh, K., Both, A., Koubarakis, M., Angelidis, I., Bereta, K., Beris, T., Bilidas, D., Ioannidis, T., Karalis, N., Lange, C., Pantazi, D., Papaloukas, C., Stamoulis, G.: *Template-based question answering over linked geospatial data*. GIR 2018

[2] <https://www.dbpedia.org/>

[3] <https://www.openstreetmap.org/>

[4] <https://gadm.org/>

GeoQA Datasets

Geospatial Datasets - GeoQuestions1089

GeoQuestions1089 [1] contains **1089** question-query-answer triples. Queries are written in **GeoSPARQL**.

- Questions crowdsourced from students
- Expanding on **GeoQuestions201**, in both complexity and scope
- Target knowledge graph:
 - **YAGO2** [2]
 - **YAGO2geo** [3] (UK, Ireland, USA and Greece)
- Currently the largest crowdsourced geospatial QA benchmark
- Follows the same question taxonomy as **GeoQuestions201**

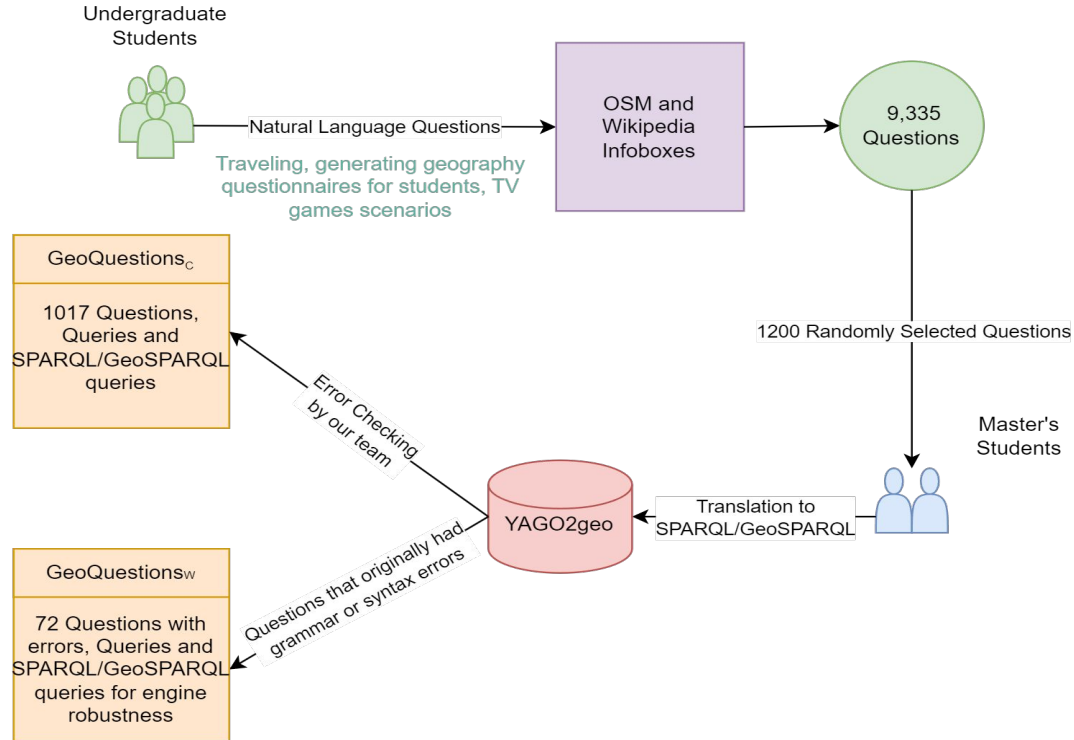
[1] Sergios-Anestis Kefalidis, Dharmen Punjani, Eleni Tsalapati, Konstantinos Plas, Mariangela Pollali, Michail Mitsios, Myrto Tsokanaridou, Manolis Koubarakis, and Pierre Maret. *Benchmarking Geospatial Question Answering Engines using the Dataset GeoQuestions1089*. ISWC 2023

[2] Hoffart, J., Suchanek, F.M., Berberich, K., Weikum, G.: *Yago2: A spatially and temporally enhanced knowledge base from wikipedia* Artif. Intell. 2013

[3] Karalis, N., Mandilaras, G.M., Koubarakis, M.: *Extending the YAGO2 knowledge graph with precise geospatial knowledge*. ISWC 2019

GeoQA Datasets

Geospatial Datasets - GeoQuestions Crowdsourcing Process



GeoQA Datasets

Geospatial Datasets - GeoQuestions Taxonomy

The questions of the datasets *GeoQuestions201* & *GeoQuestions1089* fall under the following categories:

A. Asking for a thematic or a spatial attribute of a *feature*

Example:

- Where is *Loch Goil* located?

B. Asking whether a *feature* is in a *geospatial relation* with another *feature* or *features*

Example:

- Is *Liverpool* *east of* *Ireland*?
- Is *New York City* *in* *France*?

GeoQA Datasets

Geospatial Datasets - GeoQuestions Taxonomy

The questions of the datasets *GeoQuestions201* & *GeoQuestions1089* fall under the following categories:

- C. Asking for features of a given class that are in a geospatial relation with another feature

Examples:

- Which counties border county Lincolnshire?
- Which hotels are less than two km from George Best Belfast City Airport?

- D. Asking for features of a given class that are in a geospatial relation with any features of another class

Example:

- Which churches are near castles?

- E. Asking for features of a given class that are in a geospatial relation with unspecified features of another class, and either one or all, is/are in another geospatial relation with a feature specified explicitly

Examples:

- Which churches are near a castle in Scotland?
- In Greece, which beaches are near villages?

GeoQA Datasets

Geospatial Datasets - GeoQuestions Taxonomy

The questions of the datasets *GeoQuestions201* & *GeoQuestions1089* fall under the following categories:

- F. As in categories C, D and E above, plus more thematic and/or geospatial **attributes** of the features expected as answers

Example:

- Which mountains in Scotland are **higher than 1000 meters**?

- G. Questions with **quantities** and **aggregates**

Examples:

- What is the **total area** of lakes in Oxfordshire?
- **How many** lakes are there in Epirus?
- **How many** people live in villages in Attica?

- H. Questions with **superlatives** or **comparatives**

Examples:

- Which is the **largest** island in Greece?
- Is the **largest** island in France **larger than** Crete?

GeoQA Datasets

Geospatial Datasets - GeoQuestions Taxonomy

The questions of the datasets *GeoQuestions201* & *GeoQuestions1089* fall under the following categories:

I. Questions with **quantities/aggregates**, and **superlatives/comparatives**

Examples:

- Which city in the UK has the most hospitals?
- Is the total size of lakes in Greece larger than lake Loch Lomond in Scotland?

GeoQA Datasets

Geospatial Datasets - GeoQuestions Comparison

- **GeoQuestions201**
 - mostly simple questions that can be answered with simple queries
- **GeoQuestions1089**
 - numerous complex questions that use more advanced SPARQL constructs to be answered
 - **nested queries**
 - Which is the 10th largest town in Alaska?
 - **NOT EXISTS filters**
 - Which islands don't have any lakes but have forests?
 - **arithmetic calculations**
 - How many times bigger is the Republic of Ireland than Northern Ireland?

GeoQA Datasets

Geospatial Datasets - MapQA

MapQA [1] contains **3154** question-query-answer triples. Queries are written in **SQL** and utilize **PostGIS** spatial functions.

- Dataset is generated through the use of 9 handcrafted templates and LLMs for paraphrasing.
 - Less human-effort intensive.
- Utilizes spatial information from **OSM** targeting:
 - Southern California
 - Illinois
- Resulting DB is small on disk.
- **Point geometries** means that a more limited array of spatial operations is tested.
- No taxonomy is used by the authors.

[1] Li, Zekun & Grossman, Malcolm & Eric, & Qasemi, Ehsan & Kulkarni, Mihir & Chen, Muhao & Chiang, Yao-Yi. (2025). *MapQA: Open-domain Geospatial Question Answering on Map Data*. 10.48550/arXiv.2503.07871.

GeoQA Datasets

Geospatial Datasets - MapQA Templates

	Template	Question	Answer	Category	Concepts
1	What [amenity] is adjacent to [location]?	What restaurant is adjacent to Luther Burbank Savings?	Domenico's	Geo-entity	Amenity type & Closeness
2	What are the [amenity] within 50m of [location]?	What are the bars within 50m of South Park Brewing?	Hamilton's Tavern	Geo-entity	Amenity type & Distance
3	What amenities are within a 100m radius from [location]?	What amenities are within a 100m radius of Yorba Rose?	Fuel	Attribute	Amenity type & Distance
4	What amenity is available at [location]?	What amenity is available at Port Police?	Police	Attribute	Amenity type
5	Which is closer to [location A], [location B] or [location C]?	Which is closer to ChargePoint, Chicken Maison or Ospì?	Chicken Maison	Geo-entity	Comparison & Closeness
6	What is the nearest [amenity] [direction] of [location]?	What is the nearest snack_cart west of Colours Wheelchairs?	Tomorrowland	Geo-entity	Amenity type & Direction
7	What is the nearest [amenity] to [location]?	What is the nearest clinic to East Wing?	Garden's Elderly Care	Geo-entity	Amenity type & Compare
8	Which [amenity] is nearest to the junction of [location A] and [location B]?	Which driving_school is nearest to the junction of 10th Street and 10th Street West?	Freeway Easy Traffic and Driving School	Geo-entity	Amenity type & Intersect
9	How far is [location A] from [location B]?	How far is 119th Street & Laflin from 100 Forest Place?	35141 meters	Distance	Distance

GeoQuestions Category

C

C

C

A

H

H

H

H

B

GeoQA Datasets

Geospatial Datasets - Other

The authors of [1] extract SPARQL templates from *GeoQuestions1089* creates a dataset over *OpenStreetMap* spatial data for **Singapore**.

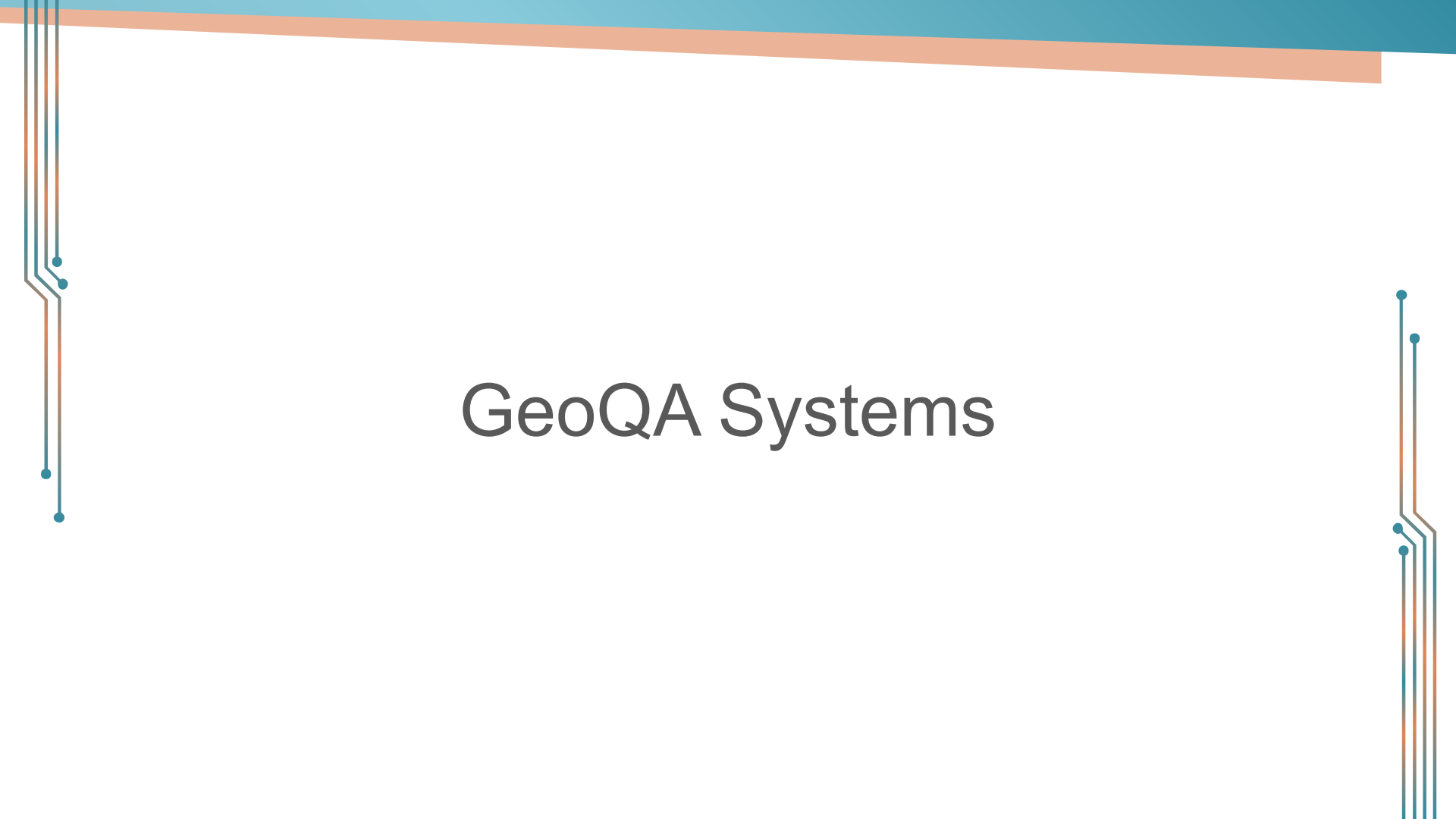
In [2], SPARQL query templates and subgraph extraction are used to generate a spatiotemporal QA dataset centered around the **administrative changes of the U.S.** over the years.

[3] is the newest dataset in our presentation. Like [2], it is also concerned with spatiotemporal QA, but it is based on **YAGO2**. It is generated based on templates.

[1] Yang, Jonghyeon, Hanme Jang, and Kiyun Yu. "Geographic Knowledge Base Question Answering over OpenStreetMap." ISPRS International Journal of Geo-Information 13.1 (2023): 10.

[2] Mitsios, Michalis, et al. "Generating a Question Answering Dataset About Geographic Changes in a Knowledge Graph." International Conference on Knowledge Engineering and Knowledge Management.

[3] Dai, Xinbang & Li, Huiying & Hu, Nan & Chen, Yongrui & Jin, Rihui & Hu, Huikang & Qi, Guilin. (2025). *Question Answering Over Spatio-temporal Knowledge Graph*. Knowledge-Based Systems. 329. 114314. 10.1016/j.knosys.2025.114314.

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GeoQA Systems

GeoQA Systems

Overview

- As we have previously discussed, GeoQA systems generally follow the Semantic Parsing paradigm and also adhere to the Entity Linking - Answer Extraction architecture.
- Early systems like the **GeoQA** family of engines relied on *heuristics* and *templates* for query generation.
 - Static systems with limited capabilities.
- These were followed by simple LLM-powered engines, like **GeoQaMap** and **GeoQA2 with GPT-4**.
 - Dynamic systems, able to answer a wider array of questions, although with a larger but manageable cost.
- The latest development is the construction of **Agents for GIS**.
 - Active R&D. Agentic systems hold a lot of promise.
- Important caveat: Most of these systems are not directly comparable, since they are either experimental with no evaluations, or introduce their own evaluation datasets.

GeoQA Systems

GeoQA Family of Engines

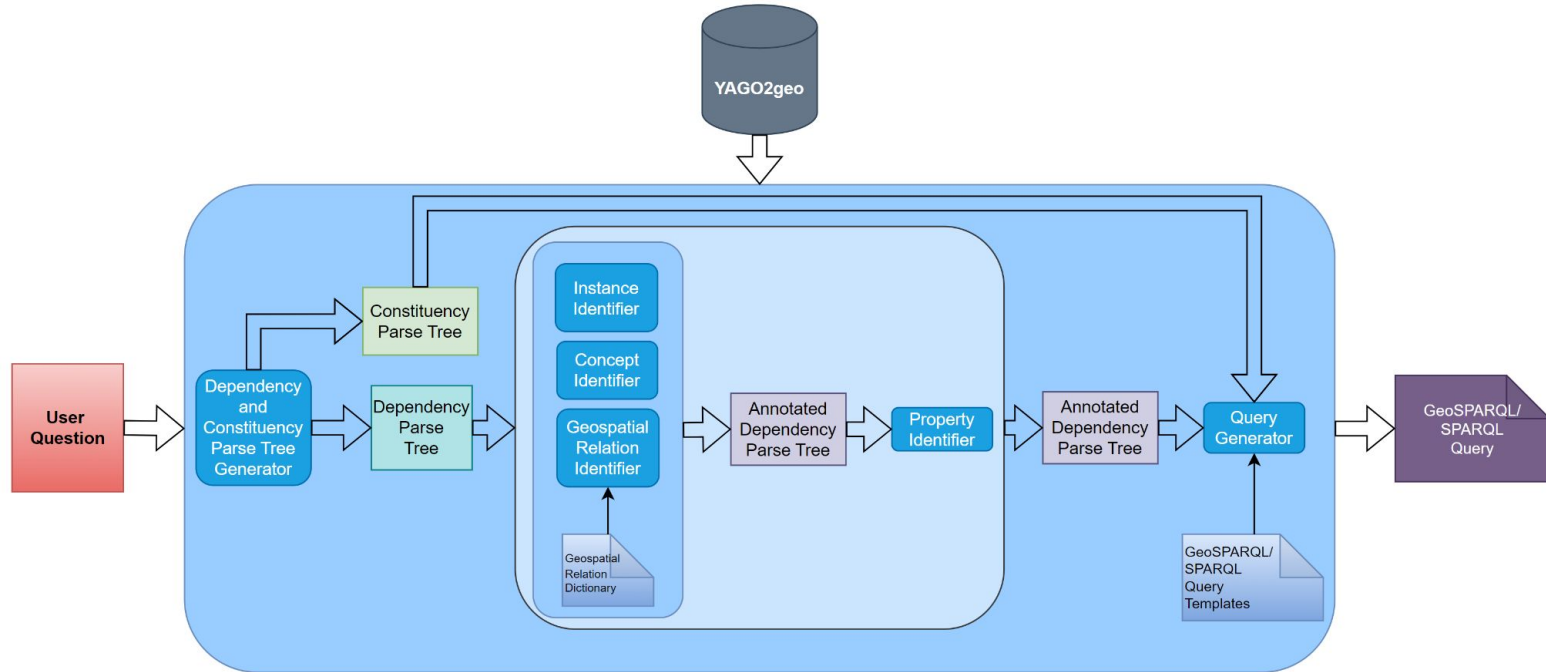
- Refers to a series of engines developed in our group. Name is a bit confusing!
- First iteration, **GeoQA** [1] released in 2018, refined in [2]:
 - Targets the **GeoQuestions201** KG
 - **Pipeline** architecture
 - Identifies *classes, entities, properties* and *geospatial relations* via a **dependency parse tree** and **n-gram matching**.
 - **9 SPARQL query templates** which are filled to generate executable queries.
 - Anything that did not fit in these templates couldn't be answered.
- Second iteration, **GeoQA2** [3], initially released in 2023, expanded in [4]:
 - Targets the **GeoQuestions1089** KG.
 - **GeoQuestions1089** is extra challenging because **YAGO2geo** is a challenging linking target.
 - Follows the same architecture but also adds the ability to **modify templated queries**, to handle more complex requests.
 - Refines all components.
- Third iteration, **GeoQA2 with GPT-4** [4] as an additional experiment to benchmark the ability of SotA LLMs on GeoSPARQL query generation.
 - **Query templates replaced by GPT-4**

[1] D. Punjani, K. Singh, A. Both, M. Koubarakis, I. Angelidis, K. Bereta, T. Beris, D. Bilidas, T. Ioannidis, N. Karalis, C. Lange, D. Pantazi, C. Papaloukas and G. Stamoulis. *Template-Based Question Answering over Linked Geospatial Data*. Geographic Information Retrieval (GIR 2018) workshop. Collocated with ACM SIGSPATIAL. Seattle, USA, 6 November, 2018

[2] Dharmen Punjani, Markos Iliakis, Theodoros Stefou, Kuldeep Singh, Andreas Both, Manolis Koubarakis, Iosif Angelidis, Konstantina Bereta, Themis Beris, Dimitris Bilidas, Theofilos Ioannidis, Nikolaos Karalis, Christoph Lange, Despina-Athanasia Pantazi, Christos Papaloukas, Georgios Stamoulis. *Template-Based Question Answering over Linked Geospatial Data*. arXiv.org

[3] Sergios-Anestis Kefalidis, Dharmen Punjani, Eleni Tsalapati, Konstantinos Plas, Mariangela Pollali, Michail Mitsios, Myrto Tsokanaridou, Manolis Koubarakis, and Pierre Maret. *Benchmarking Geospatial Question Answering Engines using the Dataset GeoQuestions1089*. The 22nd International Semantic Web Conference, Athens, Greece, 6-10 November, 2023

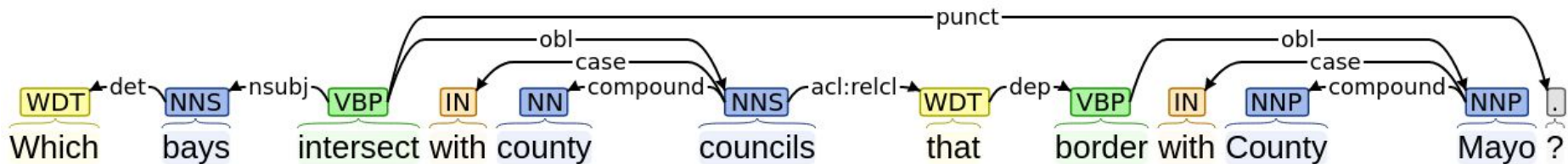
[4] Sergios-Anestis Kefalidis, Dharmen Punjani, Eleni Tsalapati, Konstantinos Plas, Maria-Aggeliki Pollali, Pierre Maret and Manolis Koubarakis. *The question answering system GeoQA2 and a new benchmark for its evaluation*. International Journal of Applied Earth Observation and Geoinformation, Volume 134, November 2024



GeoQA Systems

GeoQA Family of Engines

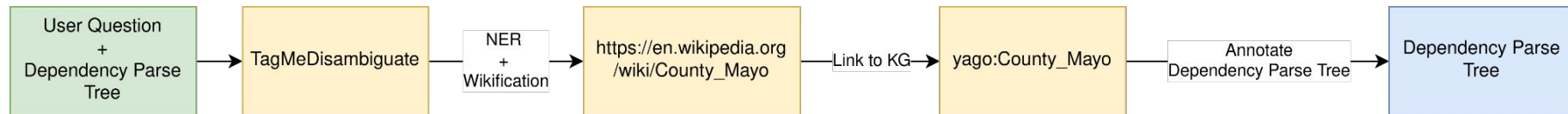
- **part-of-speech tagging**
- **dependency parse tree** generation (annotated by pipeline components)
- **constituency parse tree** generation (used by the query generator)



GeoQA Systems

GeoQA Family of Engines

- Identifying **geographic features** (instances), e.g. County Mayo, and linking to the target KG.
- How it works:



GeoQA Systems

GeoQA Family of Engines

- Which **bays** **intersect** with **county councils** that **border** with **County Mayo**?
 - Instances: **yago:County_Mayo**
 - Concepts: **y2geoo:OSI_County_Council**, **y2geoo:OSM_bay**
 - Relations: **geof:sfIntersects**, **geof:sfTouches**
 - Properties: -
- Generated Query:

```
SELECT DISTINCT ?x WHERE {  
  ?x rdf:type y2geoo:OSM_bay; geo:hasGeometry ?cGeom1. ?cGeom1 geo:asWKT ?cWKT1.  
  ?y rdf:type y2geoo:OSI_County_Council; geo:hasGeometry ?cGeom2. ?cGeom2 geo:asWKT ?cWKT2.  
  yago:County_Mayo geo:hasGeometry ?iGeom. ?iGeom geo:asWKT ?iWKT.  
  FILTER(geof:sfIntersects(?cWKT1,?cWKT2))  
  FILTER(geof:sfTouches(?cWKT2,?iWKT))  
}
```

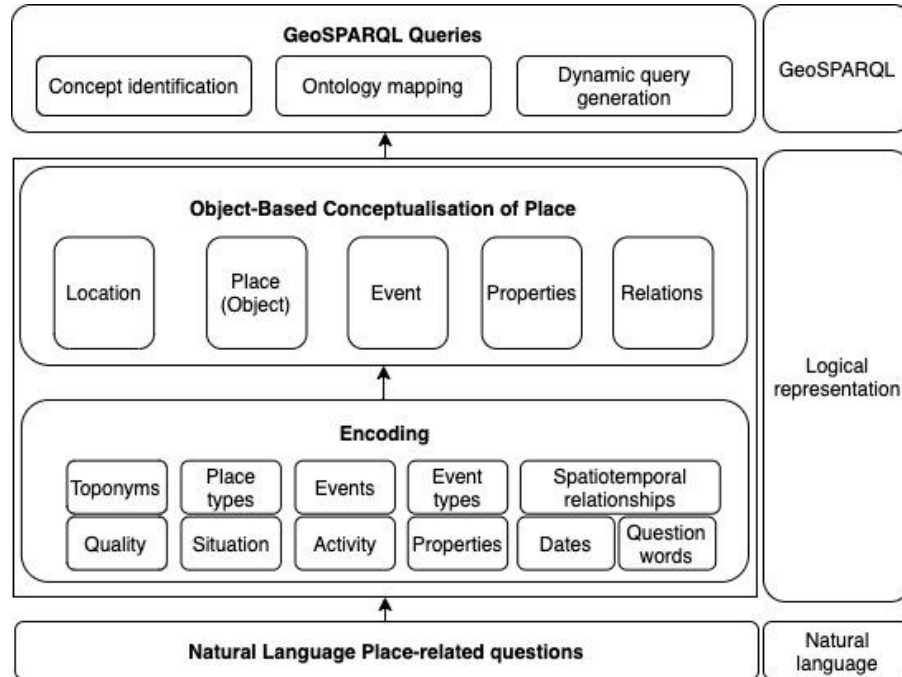
GeoQA Systems

The system of Hamzei et al

- Like **GeoQA** uses parse trees and heuristics.
- Main innovations:
 - Generates an IR (first-order logic) which is subsequently translated into SPARQL.
 - Utilizes SPARQL-blocks instead of large templates. More dynamic.
 - Utilizes some neural components for understanding question intent and entity recognition
- Weaknesses:
 - Lax entity linking results in a large amount of false positives
 - Rule-based implementation means that it is a limited system

GeoQA Systems

The system of Hamzei et al

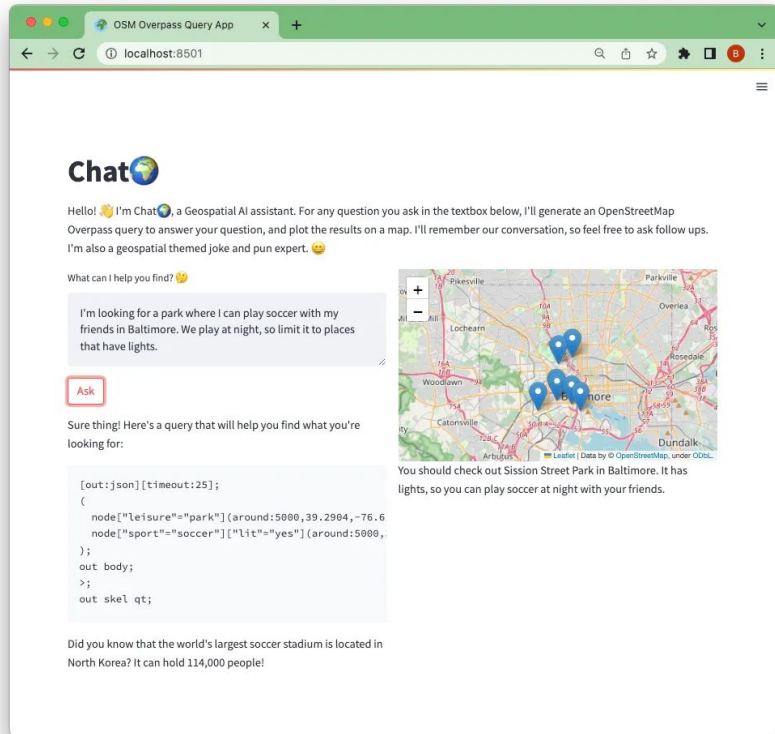


GeoQA Systems

ChatGeoPT

- Early system that utilized LLMs for dynamic GeoQA.
- Uses **OSM** via the **Overpass Querying Language**.

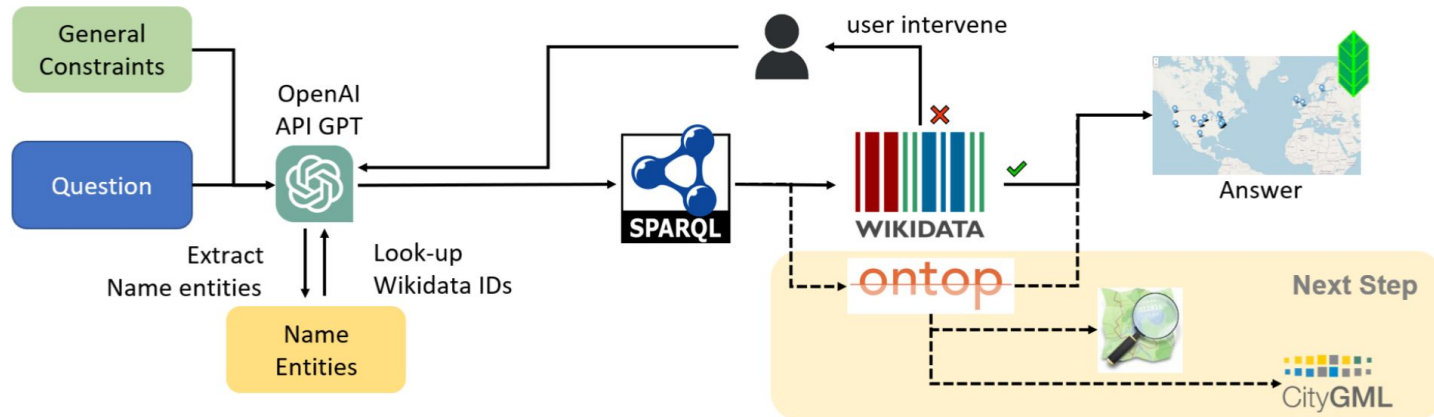
<https://github.com/earth-genome/ChatGeoPT>



GeoQA Systems

GeoQaMap

- Released in 2023.
- Targets **Wikidata**.
- **Question answering via visualization.**
- Akin to what we would now call an Agentic system.
- Experimental work, not evaluated on a benchmark.



GeoQA Systems

GeoQaMap

Prompt

all universities of Germany and China.

More requirements

The subclasses of university should also be considered.

Chatbot

all universities of Germany and China.

Name Entity: universities Name Entity: Germany Name Entity: China

- SPARQL Query: `SELECT DISTINCT ?item ?itemLabel ?coord ?description WHERE {{ ?item wdt:P31/wdt:P279* wd:Q3918 . # find instances and subclasses of university ?item wdt:P17 wd:Q183 . # filter by country Germany OPTIONAL { ?item rdfs:label ?itemLabel filter (lang(?itemLabel) = "en"). } OPTIONAL { ?item wdt:P625 ?coord . } OPTIONAL { ?item schema:description ?description filter (lang(?description) = "en"). }} UNION { ?item wdt:P31/wdt:P279* wd:Q3918 . # find instances and subclasses of university ?item wdt:P17 wd:Q148 . # filter by country China OPTIONAL { ?item rdfs:label ?itemLabel filter (lang(?itemLabel) = "en"). } OPTIONAL { ?item wdt:P625 ?coord . } OPTIONAL { ?item schema:description ?description filter (lang(?description) = "en"). }}`
- Query finished with 1156 items

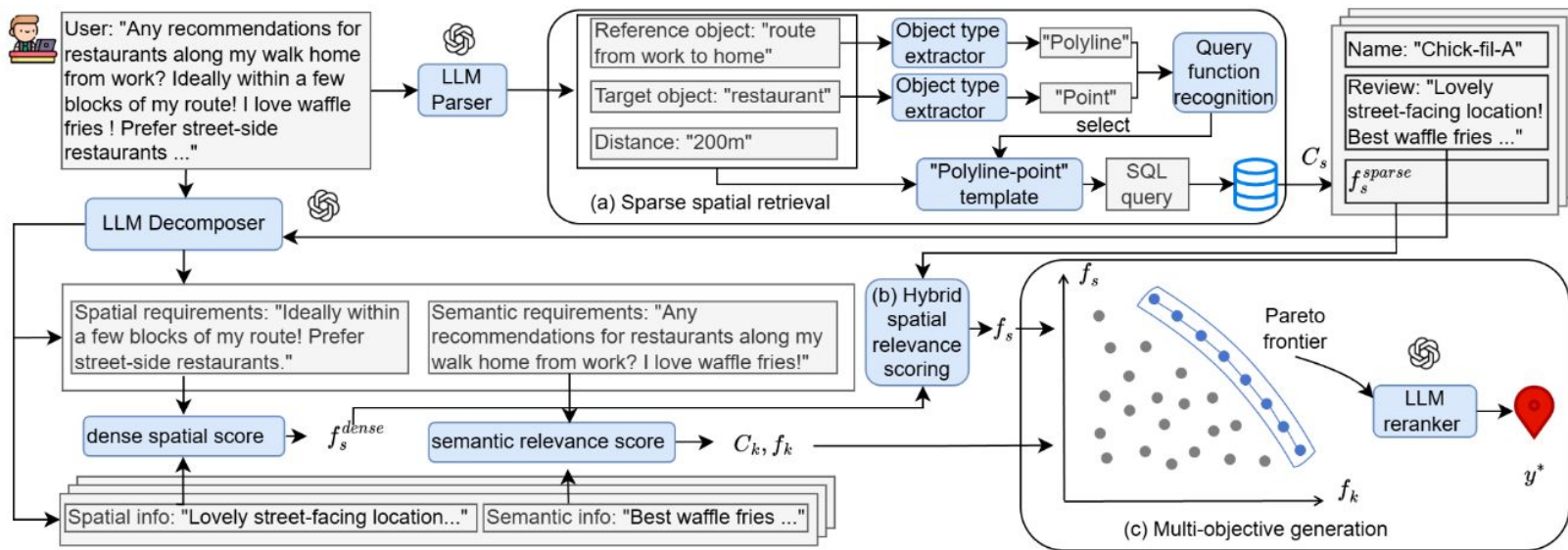


Leaflet | © OpenStreetMap contributors © CartoDB, CartoDB attributions

GeoQA Systems

Spatial-RAG

- Combines spatial operations, spatial relevance and semantic relevance to retrieve answers.
- New, novel approach.



The slide features a decorative header at the top with a blue background on the left and an orange background on the right. On the left side, there are several vertical lines in blue and orange, some ending in small dots. On the right side, there are similar vertical lines, also in blue and orange, with some ending in small dots.

Evaluation

Evaluation

Evaluation on *GeoQuestions1089*

- The engines generate a query from a natural language question.
- If the generation is successful it is run against an RDF store.
- The result is compared to the gold result included in *GeoQuestions1089*. For the generated query to be considered correct, the results must match exactly.

Evaluation

Evaluation on *GeoQuestions1089*

Table 19. Evaluation of GeoQA2 over GEOQUESTIONS_C (YAGO2geo only).

Category	Executable queries	Correct answers	Correct answers*
A	86.61%	53.52%	61.78%
B	77.61%	62.68%	80.76%
C	82.35%	48.36%	58.73%
D	54.54%	9.09%	16.67%
E	76.64%	24.81%	32.38%
F	57.14%	28.57%	50.00%
G	72.60%	36.30%	50.00%
H	65.81%	23.07%	35.06%
I	78.26%	21.73%	27.78%
Total	76.31%	40.33%	52.85%

Table 22. Evaluation of the system of Hamzei et al. (2022) over GEOQUESTIONS_C (YAGO2geo only).

Category	Executable queries	Correct answers	Correct answers*
A	100.00%	28.16%	28.16%
B	98.85%	55.22%	56.06%
C	94.11%	30.06%	31.94%
D	81.81%	4.54%	5.55%
E	93.43%	6.56%	7.03%
F	71.42%	14.28%	20.00%
G	95.89%	12.32%	12.85%
H	96.58%	33.33%	34.51%
I	95.65%	8.69%	9.09%
Total	95.41%	25.92%	27.16%

Evaluation

Evaluation on *GeoQuestions1089*

Table 19. Evaluation of GeoQA2 over GEOQUESTIONS_C (YAGO2geo only).

Category	Executable queries	Correct answers	Correct answers*
A	86.61%	53.52%	61.78%
B	77.61%	62.68%	80.76%
C	82.35%	48.36%	58.73%
D	54.54%	9.09%	16.67%
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F	57.14%	28.57%	50.00%
G	72.60%	36.30%	50.00%
H	65.81%	23.07%	35.06%
I	78.26%	21.73%	27.78%
Total	76.31%	40.33%	52.85%

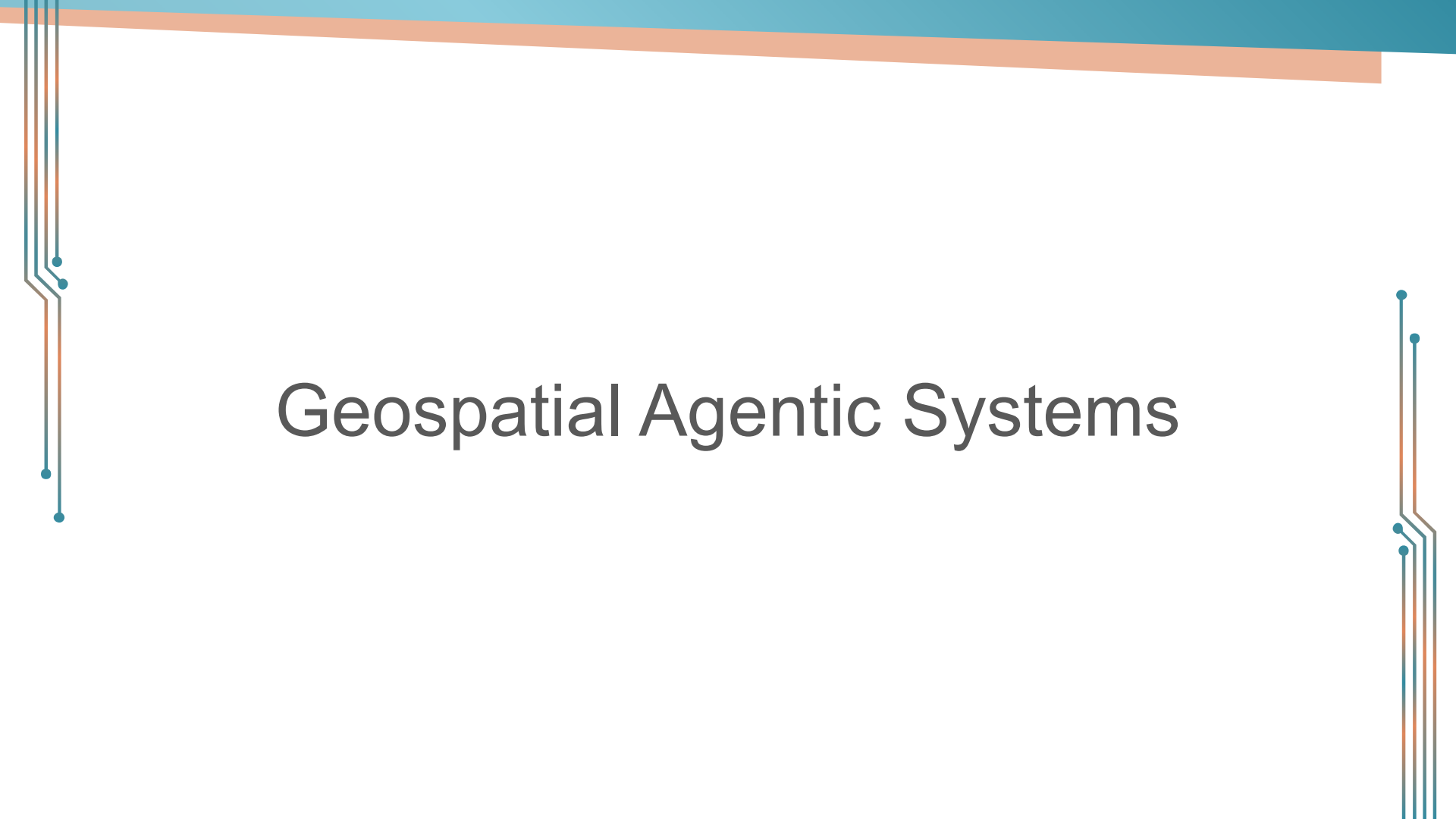
Table 25. Evaluation of GeoQA2 with GPT-4 for query generation over GEOQUESTIONS_C (YAGO2geo only).

Category	Executable queries	Correct answers	Correct answers*
A	100.00%	35.91%	35.91%
B	89.52%	62.68%	70.00%
C	92.15%	49.67%	53.90%
D	77.27%	13.63%	17.64%
E	83.21%	31.38%	37.71%
F	95.23%	42.85%	45.00%
G	84.24%	34.24%	40.65%
H	90.59%	27.35%	30.18%
I	95.65%	17.39%	18.18%
Total	89.94%	39.32%	43.72%

Evaluation

Evaluation on *GeoQuestions1089* - *GPT-3.5-turbo*

Category	Correctly answered	Partially correct/ lacking precision
Category A	15.4%	59.4%
Category B	61.4%	0%
Category C	31.3%	51.1%
Category D	23.8%	52.4%
Category E	18.8%	63%
Category F	4%	8%
Category G	6.3%	3.4%
Category H	38.7%	0%
Category I	4%	0%
Total	22.6%	26.4%



Geospatial Agentic Systems

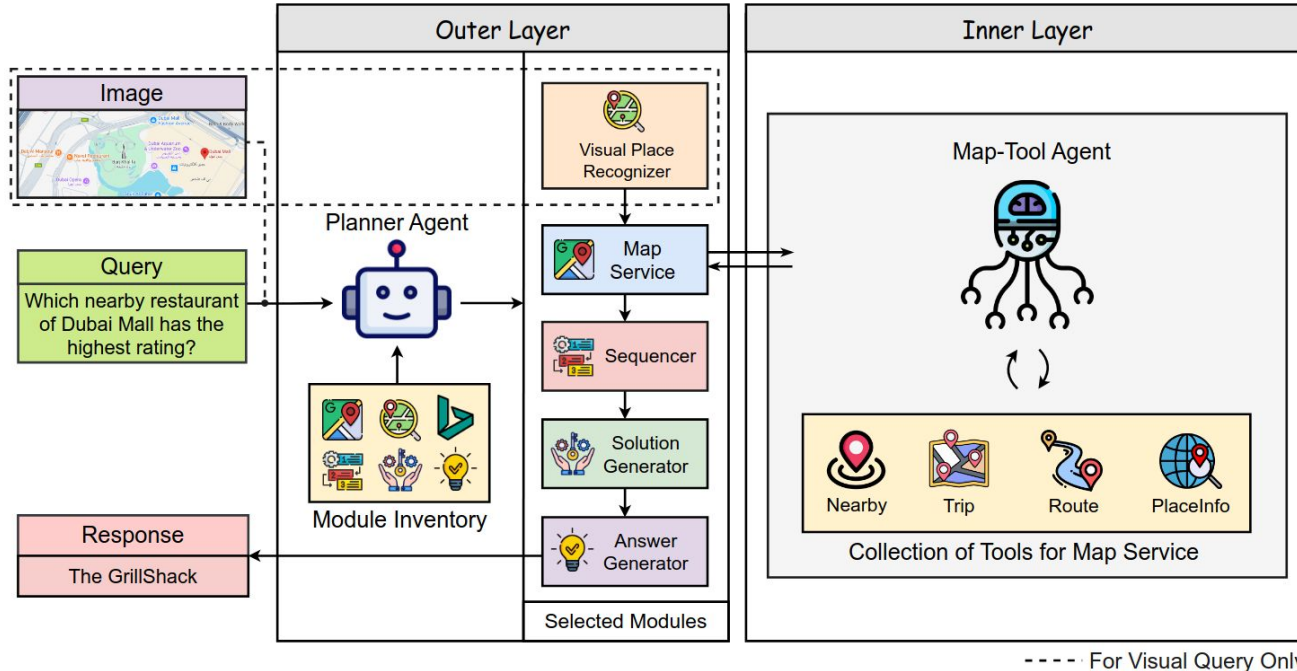
Geospatial Agentic Systems

Overview

- Agentic systems seem to be the next big step for LLM-based AI.
 - **Single-agent systems** for limited/specialized tasks.
 - **Multi-agent systems** (MAS) for complex domains and wide-ranging responsibilities.
- Tools that serve as **GIS Copilots / Assistants** are being developed to enhance data accessibility and analysis.
- These systems utilize GeoQA and KGs as tools for spatial operations and high-performance queries.

Geospatial Agentic Systems

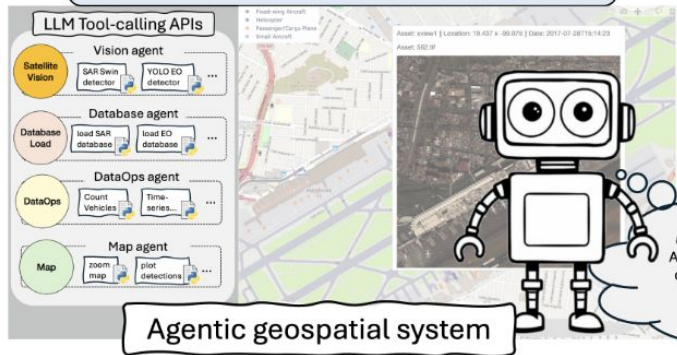
MapAgent



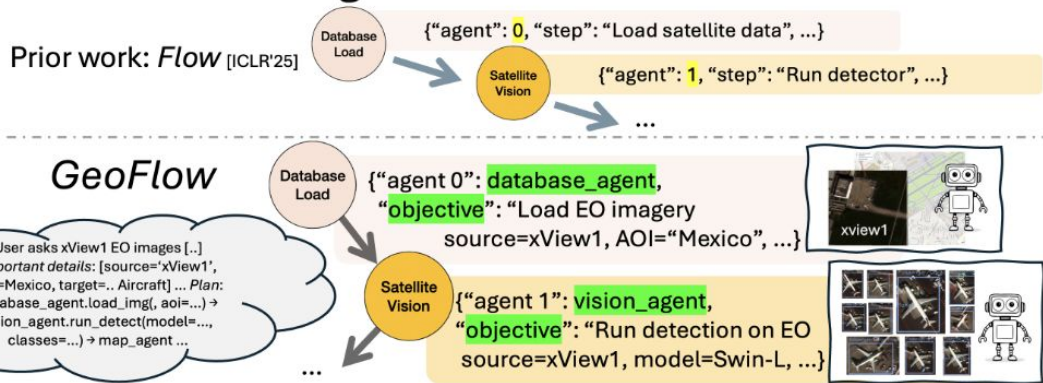
Geospatial Agentic Systems

GeoFlow

Fetch xView1 EO images from Mexico. Run Swin-L detector and plot detections for Small Aircrafts!

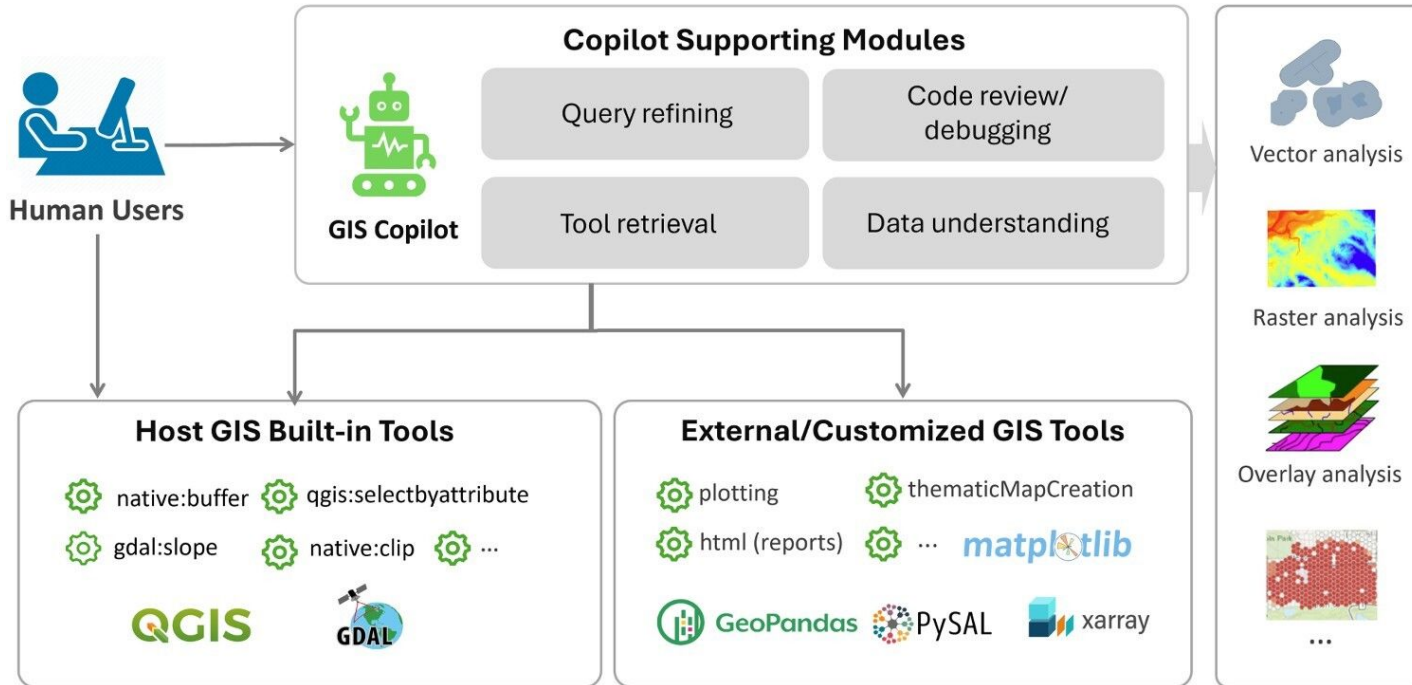


Automated Agentic Workflow Generation



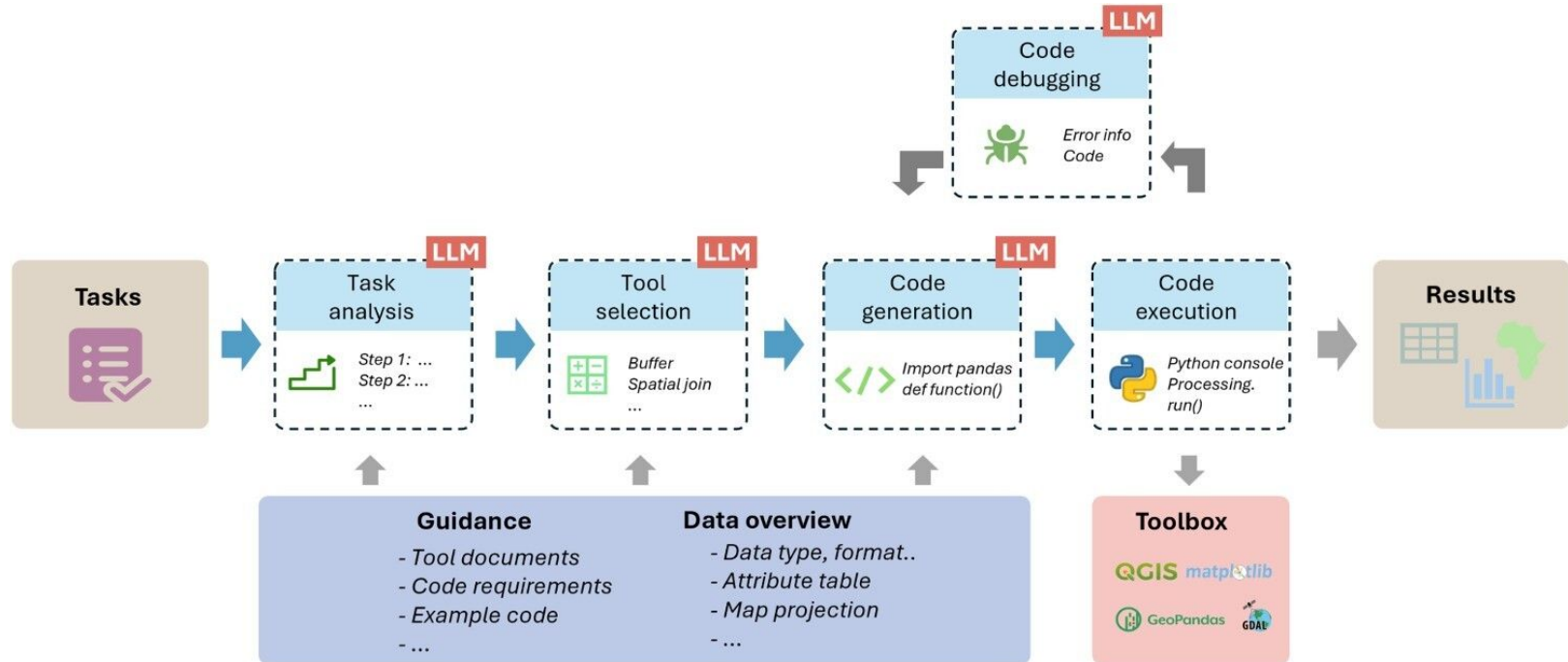
Geospatial Agentic Systems:

GIS Copilot



Geospatial Agentic Systems:

GIS Copilot



Thank you!

skefalidis@di.uoa.gr

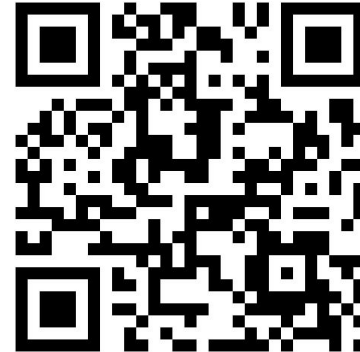
Web Pages



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